Computer-Aided Surgical Skill Assessment: Improving Surgical Care by Computer Vision

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This research statement is on the topic of computer-aided surgical skill assessment, i.e., how to automatically predict the surgeon's skill level using surgery videos. This is an interdisciplinary study of computer vision and clinical medicine. Fig. 1 gives a high-level illustration of computer-aided surgical skill assessment.

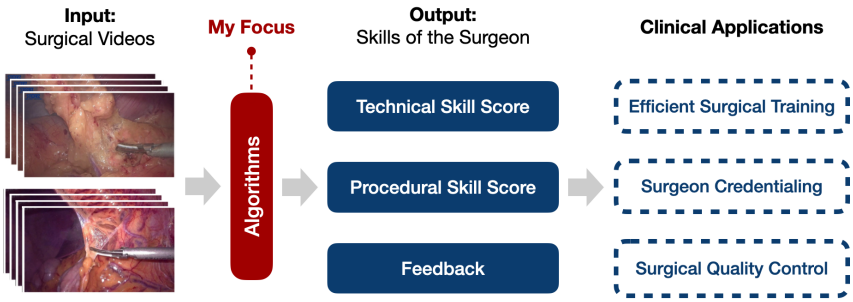

Fig. 1. Computer-aided surgical skill assessment.

1 Background

Hundreds of millions of surgical procedures are performed worldwide every year. Surgical skill assessment is crucial for the improvement of surgeons' competency during both training and practice.

Traditionally, surgical skills are manually evaluated by experts with onsite observation, which is prone to subjective biases and also labor-intensive. Thus computer-aided approaches, which provide objective and scalable skill predictions, will be essential in future surgical care.

With the prevalence of laparoscopic and robotic surgeries, a large number of surgery videos are inherently captured by the built-in cameras in surgical devices. Therefore, video-based approaches relying on computer vision techniques, which can provide efficient and non-interruptive surgical skill assessment, are likely to make a significant clinical impact and revolutionize current clinical practices.

2 Research Overview

Surgical skills can be assessed in terms of technical skills, procedural skills, and other skill proxies (Fig. 2).

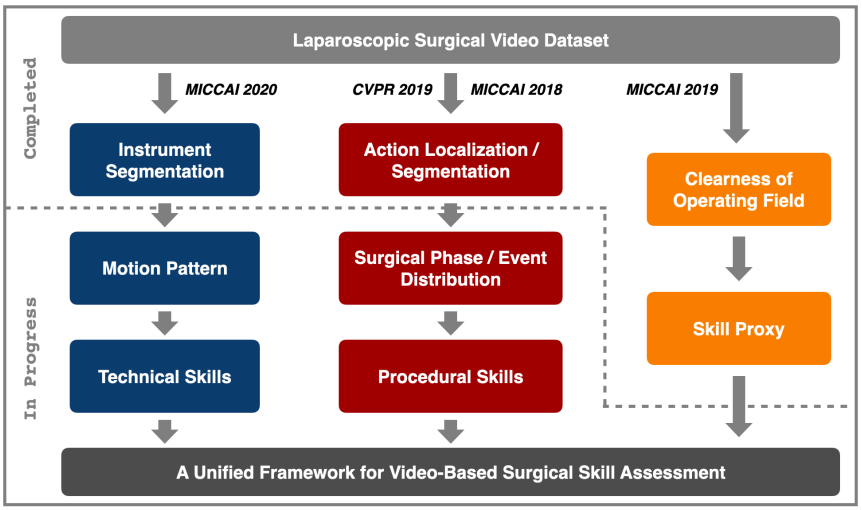

Fig. 2. Big-picture of this research statement. My research focuses on the computer-aided assessment of technical skills and procedural skills of surgeons, and the design of computational surgical skill proxies. The final goal is to improve the quality of surgical care by a unified video-based surgical skill assessment framework. My current publications are also shown in this big-picture.

2.1 Technical skills

Technical skills focus on abilities that do not necessarily depend on the surgical procedure type such as hand dexterity, motion fluency, and instrument handling proficiency [6].

The motion patterns of surgeons' hands or tools carry abundant information about technical skills. For example, an expert surgeon produces a smoother motion trajectory distributed in a smaller range. Obtaining motion patterns requires surgical instrument segmentation or tracking.

Building block technique: Surgical instrument segmentation.

2.2 Procedural skills

Procedural skills aim to evaluate the knowledge required to complete a particular surgical procedure [1], which reflect how well the surgeon understands the specifics and follows the instructions of the surgery.

Assessing procedural skills relies on the analyses of the distributions of surgical phases and events, such as dissection in anatomical planes, vessel transections, bleeding, and so on.

Building block technique: Surgical phase/action/event detection.

2.3 Skill proxies

Skill proxies are those characteristics not listed in the skill domains but highly correlated to surgical skills. We propose the clearness of operation field (COF) as a skill proxy [4]. A skillful surgeon can avoid massive bleeding and thus maintain a clean operation field so that anatomical landmarks can be recognized correctly.

3 Current Progress

3.1 Clinical surgical video dataset

Data is the basis for all computational research. In cooperation with Beijing Cancer Hospital, we collect a large clinical dataset [4], which includes fifty-seven in-vivo laparoscopic surgeries and corresponding skill scores annotated by six surgeons. This is the largest clinical dataset for surgical skill assessment so far.

3.2 Surgical phase/action detection

Surgical phase/action detection is the basis for technical skill assessment. Two of my papers have been published on surgical phase/action detection. One is on surgical gesture segmentation for simulated robotic surgeries on bench-top models using reinforcement learning [2]. The other one is on weakly-supervised temporal localization for general human activities [3]. This general computer vision work can be extended to surgical domains.

3.3 Surgical instrument segmentation

Surgical instrument segmentation is the basis for procedural skill assessment. In my recent paper on unsupervised surgical tool segmentation [5], multiple hypotheses about surgical instruments, including objectness cues, color cues and location cues, are utilized to generate learning signals for the unsupervised mining of surgical instruments. The learning signal is then refined and augmented based on the semantic correlation in the video.

3.4 Assessment via skill proxies

On our clinical dataset, we statistically identify the clearness of operation field (COF) as a good proxy for overall surgical skills, given its strong correlation with overall skills and high inter-annotator consistency [4]. Then an objective and automated framework based on the neural network is proposed to predict surgical skills through this proxy. The proposed framework achieves promising results, which are even comparable with the human performance of junior surgeons.

4 Future Plan

Future works include the dataset extension, modeling technical skills, and modeling procedural skills (Fig. 2). Besides, the burden of annotation and the model interpretability should be particularly considered.

4.1 Dataset extension

More clinical surgeries need to be collected to get more reliable models and results. In addition, the surgical phase/action and instrument bounding box/mask need to be annotated by surgeons on the current dataset.

4.2 Modeling technical skills

Given the hand/tool trajectories, technical skills can be potentially modeled via the movement range, path length, tool switch patterns, frequency, regularity, and other summary descriptors. Technical skills also depend on the type of instrument used and the type of tissue during the operation.

4.3 Modeling procedural skills

Given the detected surgical phases and actions, how to model the surgeon's procedural knowledge is an open question. The transition pattern between phases may indicate whether the operation flow is smooth. And the number of adverse events such as bleeding is strongly correlated with surgical skills.

4.4 Interpretability and feedback

The model interpretability is critical for the application of surgical skill assessment in clinical settings. Future works should focus on providing meaning skill feedback for surgeons.

4.5 Unsupervised learning

Annotations on clinical data require expensive efforts from medical experts. Future works should focus on unsupervised or weakly-supervised methods to reduce the annotation burden.

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